WEEK 4 (Mon. / Thu.)

BEFORE YOU START — WHAT TO COLLECT

Inorganic Experiment 1, Session III — catalysed oxidation

THE ONE THING THAT CANNOT BE BORROWED

0.06 g of the $RuCl_2(PPh_3)_3$ you made in Week 2. Weigh what you have left *before* the session and check it against 0.06 g, because roughly 10 mg went to ESI-MS and some was consumed by the IR. If you are short, tell the GTA at the start rather than halfway through — there is nothing to salvage once the acetone is in the flask.
Everything else must match Session II exactly: same reflux duration, same solvent batch, same technique. Bring last week's datasheet so you can copy those conditions across.

A. Glassware and apparatus

☐ 100 mL round-bottom flask — reaction
☐ 150 mL round-bottom flask — filtrate
☐ Magnetic stirrer bar (small) + retriever
☐ Water-cooled reflux condenser
☐ Rubber tubing × 2, for condenser water
☐ $CaCl_2$ guard tube + fresh $CaCl_2$ + cotton wool
☐ Retort stand + **2** clamps and bosses
☐ Hotplate stirrer + silicone oil bath + thermometer
☐ Hirsch funnel + filter paper of matching size
☐ Filter (suction) flask + vacuum trap + tubing
☐ 2 more clamps for the filtration set-up
☐ Gas adapter, for drying the oil under vacuum
☐ Measuring cylinders, 10 mL and 100 mL
☐ Micropipette **30–200 µL** + tips
☐ Glass dropper / Pasteur pipettes, clean and dry
☐ Spatula, weighing boat, NMR tube + cap + label

B. Reagents and consumables

☐ **Your $RuCl_2(PPh_3)_3$, 0.06 g** — from Week 2
☐ Dried acetone, **8 mL** (take a little extra)
☐ 1-Phenylethanol, **0.06 mL**
☐ K_2CO_3, **0.07 g**
☐ *n*-Hexane, **∼65 mL** (50 quench + 2 × 5 rinse)
☐ Celite — take enough for a *second* filtration
☐ Mesitylene, **0.07 mL** — internal standard
☐ $CDCl_3$, **0.6 mL**
☐ Silicone oil for the bath
☐ **DCM** for rinsing ruthenium-contaminated glassware

C. Shared equipment — plan around the queue

☐ Analytical balance (0.06 and 0.07 g need four decimals)
☐ **Rotary evaporator** — the bottleneck; start the reflux early
☐ Vacuum line or pump
☐ NMR submission point, and today's cut-off time

D. Personal and paperwork

☐ Own lab coat and safety goggles
☐ Full-length trousers, covered shoes, high-cut socks
☐ Gloves (nitrile provided; DCM breaks through in ∼4 min)
☐ This bench sheet and the datasheet, printed
☐ **Last week's datasheet**, for the matched conditions
☐ Calculator, permanent marker, S/N number

TWO THINGS THIS SESSION NEEDS THAT LAST WEEK DID NOT

Spare Celite. A green solid in the first filtrate is oxidised ruthenium, and the usual remedy is a second filtration. Running out of Celite at that point means either submitting a contaminated sample or improvising.
Two waste routes, located in advance. Everything containing ruthenium — filtrate, spent Celite pad, washings — goes to the **heavy-metal waste container**. The DCM rinsings go to **halogenated solvent waste**. These are different containers and both need finding before you have a beaker of green filtrate in your hand.

요약. 이번 주에 대체할 수 없는 것은 2주차에 직접 만든 $RuCl_2(PPh_3)_3$ 0.06 g이다. ESI-MS로 약 10 mg이 나갔고 IR에서도 일부 소모되었으므로, 세션 전에 남은 양을 재어 0.06 g에 미치는지 확인한다. 모자라면 아세톤을 넣기 전에 GTA에게 말해야 한다. 나머지 조건은 세션 II와 정확히 같아야 하므로 지난주 데이터시트를 가져와 그대로 옮겨 적는다. Celite는 여유 있게 챙긴다. 첫 여액에 녹색 고체가 보이면 산화된 루테늄이며 두 번째 여과가 필요한데, 그때 Celite가 없으면 대처가 어렵다. 폐기 경로가 둘로 갈린다는 점도 미리 확인한다. 루테늄이 든 것은 전부 중금속 폐기물 용기로, DCM 세척액은 할로겐 용매 폐기물로 간다.

WEEK 4 (Mon. / Thu.)

INORGANIC EXPERIMENT 1 — SESSION III

Catalytic Oxidation of 1-Phenylethanol *with* $RuCl_2(PPh_3)_3$

Venue	General & Synthetic Chemistry Lab, Block S5 Level-1	**Instrument / technique**	^{1}H NMR, use of micropipette
Time	10:00 a.m. – 4:00 p.m. (Mon. or Thu.)	**Source**	Manual v0724 + lecture notes

WHAT IS BEING MEASURED, AND SAFETY

Session II gave the background. This run adds the precatalyst and nothing else, so **the difference between the two conversions is the entire result of the experiment**. Reproduce last week's conditions and record any that you cannot.

$RuCl_2(PPh_3)_3$ is harmful and an irritant. Acetone, *n*-hexane and mesitylene are highly flammable; hexane and $CDCl_3$ are toxic and irritating. 1-Phenylethanol, acetophenone, K_2CO_3 and Celite are harmful. Fume cupboard, lab coat, goggles, gloves, covered shoes. **Everything containing ruthenium — filtrate, Celite pad, washings — goes to heavy-metal waste**; DCM rinsings to halogenated solvent waste.

Procedure

1. **8 mL** dried acetone and a stirrer bar into a 100 mL RBF.
2. By micropipette, **0.06 mL** 1-phenylethanol, with stirring.
3. **0.07 g K_2CO_3**, with stirring.
4. **0.06 g of your $RuCl_2(PPh_3)_3$**, with stirring, then fit the condenser and $CaCl_2$ guard tube *quickly. The precatalyst is air- and moisture-sensitive; every minute the flask stands open costs active catalyst.*
5. **Reflux 60 min.** Oil bath < **100 °C**. Set it once and leave it.
6. Cool to room temperature. **Cool before the next step.**
7. Add **50 mL *n*-hexane** with stirring to quench.
8. Filter through **Celite** on a Hirsch funnel; collect the filtrate. Suction flask washed and dried completely beforehand; Celite packed *with the suction already running.*
9. Transfer the filtrate to a 150 mL RBF. Rinse the suction flask with *n*-hexane (**2 × 5 mL**) and combine.
10. Remove the solvent on the rotary evaporator.
11. Fit a gas adapter; dry the oily product under vacuum **15–20 min**.
12. By micropipette, **0.07 mL mesitylene**, with stirring. *Internal standard — Appendix D.*
13. Add **0.6 mL $CDCl_3$**, transfer the whole solution into a labelled NMR tube, submit for **^{1}H NMR**.

COLOUR IS THE DIAGNOSTIC — LOG IT WITH THE TIME

The Ru(II) precatalyst is brown-black. A **deep green** mixture, filtrate or residue means oxidised, deactivated ruthenium and is the most informative observation of the session. Use the colour log on the datasheet. If a green solid appears in the first filtrate, filter through Celite a second time and record that you did.

Cool before adding hexane, or it boils out. Both flasks in the filtration must be clamped — Appendix A2.

The residue is an impure oil, so the conversion comes only from the NMR integration against mesitylene: **every transfer from the quench onwards must be quantitative**.

2-Propanol is the co-product expected only in this run and is direct evidence of hydrogen transfer — Appendix E. Its methyl doublet is expected near δ 1.2 in $CDCl_3$, so check the position before assigning; a doublet at 1.4–1.5 is far more likely leftover 1-phenylethanol.

Before you leave: ☐ NMR tube labelled and submitted ☐ Ru-bearing waste in the heavy-metal container ☐ Glassware rinsed with DCM ☐ Datasheet signed by GTA

4주차 (월 / 목)

무기 실험 1 — 세션 III

촉매 존재 하 1-phenylethanol 산화

장소	General & Synthetic Chemistry Lab, Block S5 Level-1	**장비**	^{1}H NMR, micropipette
시간	오전 10시 – 오후 4시 (월 또는 목)	**출처**	매뉴얼 v0724 + 강의노트

무엇을 측정하는가, 그리고 안전

세션 II가 배경값을 주었다. 이번 실험은 전촉매만 더할 뿐 나머지는 같으므로, **두 전환율의 차이가 이 실험의 결과 전부다.** 지난주 조건을 그대로 재현하고, 맞추지 못한 항목이 있으면 반드시 기록한다.
$RuCl_2(PPh_3)_3$는 유해하고 자극성이 있다. 아세톤, n-헥세인, 메시틸렌은 인화성이 높고 헥세인과 $CDCl_3$는 독성·자극성이 있다. 1-phenylethanol, acetophenone, K_2CO_3, Celite는 유해하다. 흄후드에서 실험복, 고글, 장갑, 발등을 덮는 신발을 착용한다. **루테늄이 든 것은 여액, Celite, 세척액까지 전부 중금속 폐기물로** 보내고, DCM 세척액은 할로겐 용매 폐기물로 보낸다.

실험 절차

1. 건조 아세톤 **8 mL**와 교반자를 100 mL 둥근바닥플라스크에 넣는다.
2. 마이크로피펫으로 1-phenylethanol **0.06 mL**를 교반하면서 가한다.
3. K_2CO_3 **0.07 g**을 교반하면서 가한다.
4. **직접 만든 $RuCl_2(PPh_3)_3$ 0.06 g**을 교반하면서 가한 뒤 냉각기와 $CaCl_2$ 건조관을 신속히 장착한다. 전촉매는 공기와 수분에 민감하여, 플라스크를 열어 둔 시간만큼 활성 촉매가 줄어든다.
5. **60분간 환류한다.** 오일배스는 **100 °C 미만**으로 두고 도중에 바꾸지 않는다.
6. 실온까지 냉각한다. **반드시 식힌 뒤 다음 단계로 넘어간다.**
7. 교반하면서 n-헥세인 **50 mL**를 가해 반응을 정지시킨다.
8. Hirsch 깔때기에 **Celite**를 깔고 감압 여과하여 여액을 받는다. 흡인 플라스크는 미리 완전히 건조해 두고, Celite는 감압을 건 상태에서 얇게 깐다.
9. 여액을 150 mL 둥근바닥플라스크로 옮기고 흡인 플라스크를 n-헥세인 **5 mL씩 2회** 헹궈 합친다.
10. 로터리 증발기로 용매를 제거한다.
11. 가스 어댑터를 연결하고 유상 생성물을 진공에서 **15–20분간** 건조한다.
12. 마이크로피펫으로 메시틸렌 **0.07 mL**를 교반하면서 가한다. 내부표준이며 부록 D를 볼 것.
13. $CDCl_3$ **0.6 mL**를 가하고 전량을 NMR 튜브에 옮겨 라벨링한 뒤 **^{1}H NMR**에 제출한다.

색이 곧 진단이다 — 시각과 함께 기록할 것

Ru(II) 전촉매는 갈흑색이다. 반응 혼합물, 여액, 잔류물이 **진한 녹색**을 띠면 루테늄이 산화되어 실활했다는 뜻이며, 이것이 이 세션에서 가장 중요한 관찰이다. 데이터시트의 색 기록표를 쓴다. 첫 여액에 녹색 고체가 보이면 Celite로 한 번 더 여과하고 그 사실을 적는다.
헥세인은 식힌 뒤에 넣는다. 뜨거우면 끓어 넘친다. 여과 시 두 플라스크를 모두 클램프로 고정한다. 부록 A2를 볼 것.
잔류물은 불순한 기름이므로 전환율은 오직 메시틸렌 대비 NMR 적분에서 나온다. **퀜칭 이후 모든 이송이 정량적이어야 한다.**
2-propanol은 이번 실험에서만 나오는 부생성물이며 수소 이동이 실제로 일어났다는 직접 증거다. 부록 E를 볼 것. 다만 $CDCl_3$에서 메틸 doublet은 δ 1.2 부근에 나오므로 위치를 확인하고 배정한다. 1.4–1.5의 doublet은 미반응 1-phenylethanol일 가능성이 훨씬 크다.

퇴실 전 확인: ☐ NMR 튜브 라벨 및 제출 ☐ 루테늄 폐기물 중금속 용기로 ☐ 유리기구 DCM 세척 ☐ 데이터시트에 GTA 서명

DATASHEET — INORGANIC EXPERIMENT 1 — WEEK 4

Session III: Catalytic Oxidation *with* $RuCl_2(PPh_3)_3$

Full name (as on student ID)	Matriculation no.	Date
Lab session ☐ Mon ☐ Thu	Fumehood no.	S/N no. assigned by lab officer

1. Reagents charged

Reagent	Nominal	Actual	Remarks
Dried acetone	8 mL		
1-Phenylethanol	0.06 mL		
K_2CO_3	0.07 g		
$RuCl_2(PPh_3)_3$ (Week 2 product)	0.06 g		
n-Hexane (quench / rinse 2×)	50 / 5 mL		
Mesitylene (internal standard)	0.07 mL		
$CDCl_3$	0.6 mL		

2. Reaction conditions

	Nominal	Actual	Remarks
Reflux start / end time	—		
Reflux duration	60 min		
Oil bath temp. / hotplate setting	$< 100\,°C$		
Vacuum drying of oil	15–20 min		

3. Colour log — record the time with each entry

On adding the catalyst		Celite pad after filtration	
During reflux		First filtrate	
After cooling		Residue after rotary evaporation	

Second filtration required? ☐ No ☐ Yes — reason: _____

4. Deviations from the Session II conditions

Solvent batch, ambient conditions, reflux duration, anything not matched

5. Other observations

_____

_____

6. NMR sample

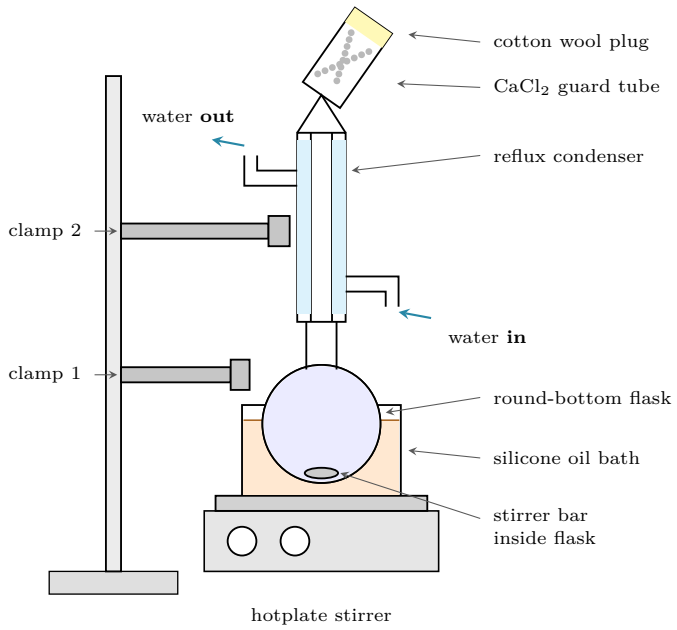
☐ Tube labelled with name, session and S/N no. ☐ Submitted S/N no. _____

To be completed at the end of the session		
Student signature	GTA name	GTA signature

Raw data and observations must be signed by the GTA before leaving the laboratory. This datasheet is submitted as an appendix to the report.

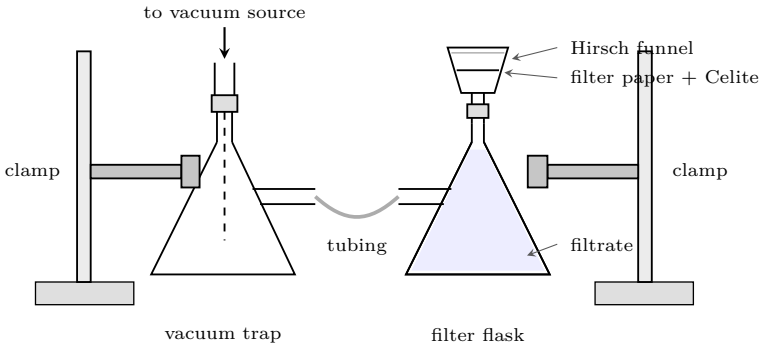
Appendix A — Apparatus Setup

A1. Moisture-free reflux assembly



Three things this drawing is meant to fix. Cooling water enters at the *bottom* and leaves at the *top*, so the jacket stays full; reversing it leaves an air pocket and the condenser stops working. **Clamp 2** holds the condenser and **clamp 1** holds the flask neck, so that neither rests its weight on the oil bath. Both grip the *necks*, and they are snug rather than forced: glass that is being heated expands, and a rigid clamp bearing on it will crack it. The grip must still be secure, because a condenser that slips lets cooling water run into the hot oil. The guard tube is packed with $CaCl_2$ and plugged with cotton wool; it excludes moisture while still venting, so it must never be sealed.

A2. Suction filtration with a vacuum trap



Both flasks must be clamped. A filter flask under vacuum tips easily once tubing is attached, and thick-walled glass under reduced pressure fails violently. The trap sits between the flask and the vacuum line so that liquid drawn over cannot reach the pump; it must be *empty* before use, and its inner tube must extend below the side arm or it will not trap anything. For Session I the funnel carries filter paper only; for Sessions II and III a thin layer of Celite is packed on top of the paper *with the suction already running*, so the pad does not lift.

요약. 냉각수는 아래로 들어가 위로 나가야 재킷이 물로 가득 찬다. 반대로 연결하면 공기층이 생겨 냉각이 되지 않는다. 클램프는 플라스크 목과 냉각기를 각각 잡아 무게가 오일배스에 실리지 않게 하되, 가열되는 유리를 힘으로 조이지 않는다. 유리가 팽창하면서 깨진다. 그렇다고 느슨하면 냉각기가 밀려 물이 기름으로 흘러들므로 알맞게 물리는 것이 요령이다. 건조관은 수분을 막되 압력은 빠져나가야 하므로 절대 밀봉하지 않는다. 감압 여과에서는 여과 플라스크와 트랩을 모두 클램프로 고정하며, 트랩의 안쪽 관은 반드시 곁가지보다 아래까지 내려와 있어야 한다.

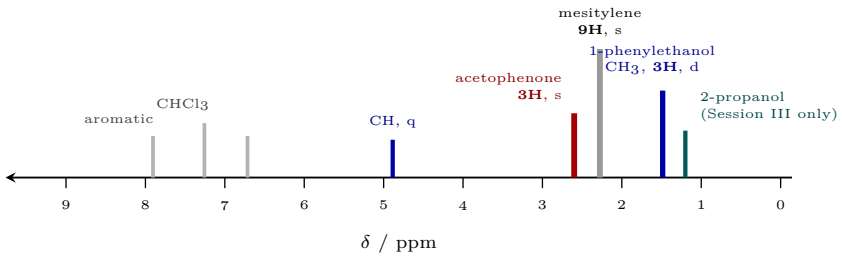
Appendix D — Quantitative ^{1}H NMR with an Internal Standard

D1. Why the yield cannot be weighed

After the rotary evaporator you hold an oily residue of unreacted 1-phenylethanol, any acetophenone formed, traces of solvent and, in Session III, ruthenium residues. Weighing it measures the mixture, not the product. The conversion must therefore come from the ^{1}H NMR spectrum — but an integral is only a *ratio*, and becomes a molar amount only when something of known amount shares the tube. That is what mesitylene is for.

A usable standard must be inert towards everything in the tube, must not overlap the signals being measured, must be added in an accurately known amount, and must survive the sample preparation. Mesitylene qualifies on all four counts: an unreactive arene whose nine equivalent methyl protons give one tall singlet in an otherwise clear region, delivered as 0.07 mL by micropipette, a known 0.503 mmol. This is also why it goes in *after* the solvent is stripped — add it earlier and the rotary evaporator removes an unknown fraction, and the known amount is no longer known.

D2. Where the signals fall



The two signals that matter are well separated: **mesitylene CH_3 near δ 2.3 (9H, singlet)** and **acetophenone $COCH_3$ near δ 2.6 (3H, singlet)**. Unreacted 1-phenylethanol is identified by its methyl *doublet* near δ 1.5 and its methine quartet near δ 4.9. In Session III only, 2-propanol appears as the hydrogen-transfer co-product; its methyl doublet is expected near δ 1.2 in $CDCl_3$, so a doublet at 1.4–1.5 is far more likely to be leftover 1-phenylethanol — check the multiplet positions before assigning it.

D3. The calculation

An integral is proportional to the number of protons, so dividing each integral by the number of protons giving rise to it puts both species on the same molar footing:

$$n_{\text{analyte}} = \frac{I_{\text{analyte}}/H_{\text{analyte}}}{I_{\text{standard}}/H_{\text{standard}}} \times n_{\text{standard}}$$

WORKED EXAMPLE — SUBSTITUTE YOUR OWN INTEGRALS

Set the mesitylene singlet to 9.00 when you integrate. Suppose the acetophenone singlet then integrates to 0.12. With $n_{\text{standard}} = 0.503$ mmol,

$$n_{\text{acetophenone}} = \frac{0.12/3}{9.00/9} \times 0.503 = 0.0201 \text{ mmol},$$

which against the 0.4975 mmol of 1-phenylethanol charged is a conversion of **4.0%**. The same arithmetic with an integral of 0.28 gives 0.0470 mmol and **9.4%**. The difference between the two sessions *is* the result of the experiment.

D4. Things that quietly ruin the number

Material lost in transfer	The standard is referenced against the *whole* residue. Anything left in the suction flask, on the Celite or in the rotary evaporator flask lowers the apparent conversion.
Standard added too early	Mesitylene largely survives the rotary evaporator, but not reliably. Add it afterwards, as the manual specifies.
Incomplete dissolution	Undissolved residue is invisible to NMR. Swirl until the $CDCl_3$ solution is homogeneous.
Overlapping solvent peaks	Residual $CHCl_3$ 7.26, water ~1.56, acetone ~2.17, hexane 0.88 and 1.26. Acetone sits uncomfortably close to the mesitylene singlet.
Careless integration	Integrate over a defined region with flat baseline on both sides, and integrate identically in both sessions or the comparison is meaningless.

Submission requirement. At least one spectrum per session must be presented as a **full spectrum from 0 to 10 ppm with every signal integrated** and the file name visible on it. Integrating only the region of interest will lose marks even if the number is right.

요약. 회수한 유상 잔류물은 불순물이라 무게로 수율을 낼 수 없다. NMR 적분값은 비율일 뿐이므로, 같은 튜브에 양을 정확히 아는 물질이 있어야 몰수로 환산된다. 메시틸렌이 그 역할을 하며, 메틸 9H가 한 봉우리로 깨끗한 영역에 나오기 때문에 적합하다. 용매를 다 날린 뒤에 넣는 이유는 그 전에 넣으면 로터리 증발기에서 얼마가 날아갔는지 알 수 없어 '아는 양'이 아니게 되기 때문이다. 계산은 각 적분값을 해당 수소 수로 나눈 뒤 비를 취하고 표준의 몰수를 곱하면 된다. 퀜칭 이후 모든 이송이 정량적이어야 하며, 두 세션의 적분 방식이 같아야 비교가 성립한다.

Appendix E — What the Catalyst Is Actually Doing

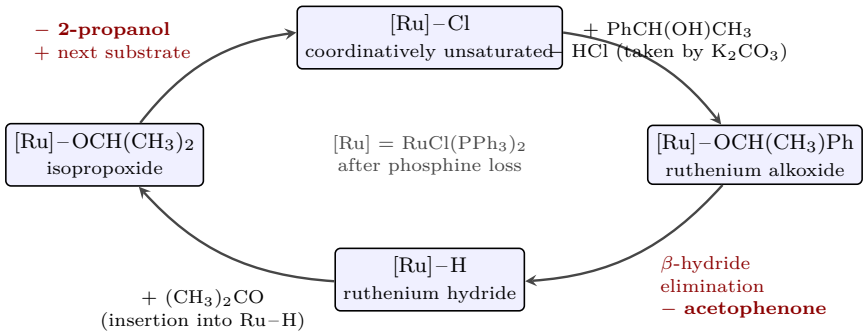
E1. Transfer dehydrogenation, not oxidation by air

The net reaction moves two hydrogen atoms from the alcohol to acetone. Nothing is oxidised by dioxygen, which is why the yield in Session II was so small:

$$\mathrm{PhCH(OH)CH_3} + \mathrm{(CH_3)_2CO} \xrightarrow{\mathrm{RuCl_2(PPh_3)_3,\ K_2CO_3}} \mathrm{PhC(O)CH_3} + \mathrm{(CH_3)_2CHOH}$$

Acetone is both the solvent and the hydrogen acceptor. Its enormous excess is what pulls an otherwise unfavourable equilibrium towards acetophenone, and the 2-propanol it becomes is the fingerprint of the mechanism: it appears only when the catalyst is working.

E2. The cycle



$\mathrm{RuCl_2(PPh_3)_3}$ is a **precatalyst**, not the catalyst. It must first shed a phosphine to open a coordination site — the same lability the ESI-MS spectrum reports on in Week 2. The base then lets the alcohol bind as an alkoxide, and the key step is **β-hydride elimination**: a hydrogen on the carbon adjacent to the bound oxygen transfers to ruthenium, and the C–O bond becomes the C=O of acetophenone. The metal hydride then hands that hydrogen on to acetone, which leaves as 2-propanol and regenerates the alkoxide-forming species. Both published mechanisms for this system — Bäckvall's of 1996 and the dihydride route of Szabó and Bäckvall in 1999 — differ in the resting state but share this elimination step.

E3. Turnover — the numbers to put in the report

Conversion alone does not describe a catalyst. Two further quantities do:

$$\mathrm{TON} = \frac{n_{\mathrm{acetophenone}}}{n_{\mathrm{catalyst}}} \qquad \mathrm{TOF} = \frac{\mathrm{TON}}{t}$$

WORK IT OUT WITH YOUR OWN NUMBERS — AND READ WHAT IT MEANS

With 0.06 g of $\mathrm{RuCl_2(PPh_3)_3}$ (MW 958.83) you charge 6.3×10^{-5} mol of ruthenium. If the NMR gives 4.7×10^{-5} mol of acetophenone after 60 min, then TON ≈ 0.75 and TOF $\approx 1.2 \times 10^{-2}$ $\mathrm{min^{-1}}$.

A TON below one means the average ruthenium centre did not complete a single turnover. Stated plainly, under these conditions the compound is behaving less like a catalyst than like a stoichiometric reagent that has partly deactivated. This is not a failed experiment — it is the honest result, and saying so is better than calling a sub-unity TON evidence of catalysis. The comparison that carries the report is the ratio between Sessions II and III, not the absolute conversion.

E4. Why it stops

The ruthenium(II) precatalyst is brown-black. **A deep green colour at any stage means oxidised ruthenium**, and oxidised ruthenium is out of the cycle. The likely entry points are dioxygen admitted while the condenser is being fitted — which is why the manual says to connect it *quickly* — and moisture, since water competes with the alcohol for the vacant site. A green filtrate should be filtered through Celite a second time before the NMR sample is made up, and the observation recorded with the time it appeared.

Two further checks worth making on the spectrum. If **2-propanol is absent** while acetophenone is present, hydrogen transfer to acetone is not the operating pathway and the acetophenone is background, as in Session II. If **free $\mathbf{PPh_3}$ or $\mathbf{OPPh_3}$** appears in the aromatic region, phosphine has been shed and oxidised, which is consistent with the deactivation above.

요약. 이 반응은 공기 산화가 아니라 이동 탈수소화이며, 아세톤이 용매이자 수소 수용체 역할을 한다. $\mathrm{RuCl_2(PPh_3)_3}$는 촉매 자체가 아니라 전촉매여서 포스핀 하나를 잃고 배위 자리를 열어야 시작되는데, 이는 2주차 ESI-MS가 보여주는 바로 그 리간드 불안정성이다. 핵심 단계는 β-수소 제거이고, 생성된 금속 수소화물이 아세톤에 수소를 넘기면 2-propanol이 떨어져 나오면서 사이클이 닫힌다. 따라서 2-propanol의 유무가 이 메커니즘이 실제로 돌았는지를 가르는 지표다. TON이 1보다 작으면 평균적으로 루테늄 한 중심이 한 바퀴도 돌지 못했다는 뜻이며, 이를 촉매 작용의 증거라고 쓰기보다 있는 그대로 적는 편이 낫다. 보고서의 결론을 지탱하는 것은 절대 전환율이 아니라 세션 II와 III의 비교다. 진한 녹색은 산화된 루테늄, 즉 실활을 뜻하므로 관찰 시각과 함께 기록한다.